

RITVIK RAO

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EDUCATION

Ph.D. in Computer Science 2022 – Dec. 2027 (expected)
University of Illinois Urbana-Champaign Urbana, IL

- Advisor: Prof. Laxmikant V. Kalé, Parallel Programming Laboratory
- Thesis topic: optimizations for large-scale fine-grained irregular applications

B.S. in Computer Engineering and Computer Science Dec. 2021
Northeastern University. Minor in Mathematics; GPA 3.91/4.0 Boston, MA

RESEARCH EXPERIENCE

Graduate Researcher 2022 – present
Parallel Programming Laboratory, University of Illinois Urbana-Champaign Urbana, IL

- Primary designer of **Reconverse**, a scheduling and communication layer replacing Charm++'s 30-year-old Converse, with lottery-queue scheduling, queue registration, and an LCI machine layer; up to $2\times$ speedups on large-scale applications
- Developed ACIC, an adaptive asynchronous shortest-path algorithm for high-diameter graphs with tens of billions of vertices, up to two orders of magnitude faster than existing distributed graph frameworks
- Accelerated friends-of-friends cluster finding for cosmological simulations by combining tree-traversal and union-find libraries with fine-grained optimizations for highly imbalanced datasets
- Implemented the Ray Core API on top of **Charm4Py**, letting unmodified Ray programs use dynamic load balancing by mapping Ray actors to Charm4Py chares; $3\text{--}4\times$ speedups on Ray benchmarks over the standard Ray runtime
- Built a library of graph algorithms for CharmTyles, an interactive computing interface to Charm++ aimed at data scientists

Undergraduate Researcher, Julia on AMD GPUs June 2021 – June 2022
NUCAR Lab, Northeastern University Boston, MA

- Designed an interface that translates Julia GPU code to HIP for AMD's ROCm platform, allowing Oceananigans, an ocean climate model written in Julia, to run on AMD GPUs

Pi-Tiles Jan. 2019 – Apr. 2019
NUCAR Lab, Northeastern University Boston, MA

- Implemented blocked matrix multiplication in MPI across a cluster of Raspberry Pis, $10\times$ faster than the naive method

INDUSTRY AND LABORATORY EXPERIENCE

Ph.D. Extern May 2025 – Aug. 2025
IBM Research

- Designed a portable collective communication library for AI workloads (all-gather, all-reduce, reduce-scatter), building on and contributing to the open-source UCCL library for moving large tensors between devices
- Combined UCCL with Charm++ so that collectives tolerate accelerator failures and adapt to any cluster's hardware and network topology; demonstrated on a distributed tensor multiplication written in Charm++

Ph.D. Intern May 2023 – Aug. 2023

Pacific Northwest National Laboratory

- Developed a minimal-communication asynchronous algorithm for distributed reinforcement learning training on GPUs, built with RLlib on the Ray API, and applied it to speed up training for the influence maximization problem

Control Systems Engineering Co-op

Jan. 2021 – June 2021

Nuvera Fuel Cells

- Developed Simulink control software for a hydrogen fuel cell engine, raising power output from 45 to 60 kW and lowering the required starting temperature from 0 °C to –30 °C
- Ported fuel cell stack partial-derivative calculations from MATLAB to C, and ran lab tests analyzing performance data to tune the control code

GPU Software Engineering Co-op

Jan. 2020 – June 2020

Advanced Micro Devices (AMD)

- Designed a tool that instruments GPU shader programs during compilation to record performance metrics at run time
- Worked with the shader compiler team to define project goals and specifications, and surveyed similar tools from other companies

PUBLICATIONS

Journal Articles

- [J1] Joseph Schuchart, Patrick Diehl, Michael Bauer, Aurelien Bouteiller, Gregor Daiss, Engin Kayraklioglu, Shreyas Khandekar, Thomas Herault, John Holmen, **Ritvik Rao**, Alexander Strack, Elliott Slaughter, Jennifer Spinti, Jeremy Thornock, Alex Aiken, Olivier Aumage, Martin Berzins, George Bosilca, Bradford Chamberlain, Hartmut Kaiser, and Laxmikant V. Kalé. “A Survey of Distributed Asynchronous Many-Task Models and Their Applications.” *ACM Computing Surveys*, 2026. DOI 10.1145/3840389

Conference and Workshop Papers

- [C1] **Ritvik Rao**, Jiakun Yan, Aditya Bhosale, and Laxmikant V. Kalé. “Reconverse: A New Scheduling Library for Multithreaded Task-Based Runtime Systems.” In *9th Annual Parallel Applications Workshop, Alternatives To MPI+X (PAW-ATM), held with SC26*, 2026. To appear.
- [C2] **Ritvik Rao** and Laxmikant V. Kalé. “Efficient Cosmological Friends-of-Friends Computation with Fine-Grained Optimizations.” In *16th IEEE/ACM Workshop on Irregular Applications: Architectures and Algorithms (IA³), held with SC26*, 2026. To appear.
- [C3] Aditya Bhosale, Anant Jain, Shourya Goel, **Ritvik Rao**, Peddoju Sateesh Kumar, and Laxmikant V. Kalé. “Efficient and Portable Support for Overdecomposition on Distributed GPU Platforms.” In *9th Annual Parallel Applications Workshop, Alternatives To MPI+X (PAW-ATM), held with SC26*, 2026. To appear.
- [C4] **Ritvik Rao**, Kavitha Chandrasekar, and Laxmikant V. Kalé. “An Adaptive Asynchronous Approach for the Single-Source Shortest Paths Problem.” In *SC24-W: Workshops of the International Conference for High Performance Computing, Networking, Storage and Analysis (IA³)*, pp. 697–702, 2024.

TALKS AND TUTORIALS

- Tutorial (co-presenter): “Charm4Py: A Programming Model for Distributed Adaptive Python.” *IPDPS 2026*, New Orleans, LA. May 2026
- “Overview of Reconverse.” *JLESC 2026*, Jülich, Germany. May 2026
- “Runtime Adaptivity for CSE and Machine Learning with Charm4Py, CharmTyles, and CharmNumerics.” *JLESC 2026*, Jülich, Germany. May 2026

“An Adaptive Asynchronous Approach for the Single-Source Shortest Paths Problem.” *IA³* at SC24, Atlanta, GA. Nov. 2024

TEACHING

Teaching Assistant, CS 173: Discrete Structures. University of Illinois Urbana-Champaign	Fall 2023
Teaching Assistant, CS 4120: Natural Language Processing. Northeastern University	Fall 2021
Teaching Assistant, CS 1800: Discrete Structures (3 semesters). Northeastern University	2019 – 2021

FUNDING

Research Assistant, Theoretical and Computational Biophysics Group (TCBG), Beckman Institute, University of Illinois Urbana-Champaign	Fall 2026 – present
Research Assistant, IBM-Illinois Discovery Accelerator Institute (IIDAI)	Spring 2024 – Spring 2026
Teaching Assistant, CS 173: Discrete Structures, University of Illinois Urbana-Champaign	Fall 2023
First-year fellowship, awarded to all incoming UIUC CS Ph.D. students	Fall 2022 – Spring 2023

HONORS AND AWARDS

Northeastern National Merit Scholarship, Northeastern University	
Dean’s List, Northeastern University (Fall 2018; Spring, Summer I, and Fall 2019; Fall 2020)	2018 – 2020

SERVICE

Student Volunteer, SC26, Chicago, IL	Nov. 2026
Student Volunteer, SC24, Atlanta, GA	Nov. 2024

SKILLS

Languages: C++, C, Python, Java

Parallel and GPU programming: Charm++, Charm4Py, MPI, OpenMP, CUDA, ROCm/HIP, LCI, Ray/RLlib, UCCL

Tools: Slurm, Projections, Git, MATLAB/Simulink, $\text{\LaTeX}$